

TANGENT SPACE (AT point p in S)

Now consider a function f over the curve c .

$$f \circ c : [a, b] \rightarrow \mathbb{R}.$$

Let Cf be the derivative of this function along the curve.

$$\begin{aligned} Cf &= \frac{d f \circ c}{dt} = \frac{d f(\theta(t))}{dt} \\ &= \sum_{i=1}^n \frac{d\theta^i}{dt} \frac{\partial f}{\partial \theta^i} \end{aligned}$$

Cf is the directional derivative associated with the curve c .

↳ The rate of change of the function in direction of the tangent to the curve c at $\theta(t)$.

TANGENT SPACE (AT point p in S)

One can think of the operator C in

$$Cf = \sum_{i=1}^n \frac{d\theta^i}{dt} \frac{\partial f(\theta(t))}{\partial \theta^i}$$

as the sum of vectors $\frac{\partial}{\partial \theta^i}$ scaled

by their components $\frac{d\theta^i}{dt}$.

$\frac{\partial}{\partial \theta^i}$ is the rate of change w.r.t the

i th coordinate of the curve $\theta(t)$.

We want to:

- Create a basis at point $p \in S$
that spans the tangent space T_p at p .
- Tangent space is the set of all
tangent vectors of curves through p .

Construct n curves through point $p_0 \in S$.
(indexed $(i), i=1, \dots, n$)

In the coordinate system we
created using the homeomorphism
 ϕ ,
 p_0 is $\theta_0 = (\theta_0^1, \theta_0^2, \dots, \theta_0^n)$.

$$\text{Curve } \theta_{(1)}(t) = (\theta_0^1 + t, \theta_0^2, \dots, \theta_0^n)$$

$$\frac{d}{dt} f(\theta_{(1)}(t)) = \frac{\partial f}{\partial \theta^1}$$

$$\vdots$$
$$\text{Curve } \theta_{(i)}(t) = (\theta_0^1, \theta_0^2, \dots, \theta_0^i + t, \dots, \theta_0^n)$$

$$\frac{d}{dt} f(\theta_{(i)}(t)) = \frac{\partial f}{\partial \theta^i}, \quad i=1, 2, \dots, n.$$

FACT:

$\partial_i \triangleq \frac{\partial}{\partial x^i}$, $i=1, 2, \dots, n$ constitute a basis.

Any directional derivative operator A can be written as:

$$\left. \begin{aligned} A &= \sum_{i=1}^n A^i \frac{\partial}{\partial x^i} \\ &= \sum_{i=1}^n A^i \partial_i \end{aligned} \right\} \begin{array}{l} A^i \text{ is the} \\ \text{component of} \\ A \text{ along} \\ \frac{\partial}{\partial x^i} \end{array}$$

We have the vector space

$$T_p = \left\{ A : A = \sum_{i=1}^n A^i \partial_i \right\}$$

Ⓡ Every linear combination of ∂_i is a directional derivative at $p \in M$. Recall $\partial(p) = 0$.

Ⓡ Think of T_p above as T_p dot product. We can write:

$$\nabla_p^T \begin{bmatrix} A_i \end{bmatrix}$$

INNER PRODUCT $\langle A, B \rangle$

$$\langle A, B \rangle = \left\langle \sum_{i=1}^n A^i \partial_i, \sum_{j=1}^n B^j \partial_j \right\rangle$$

$$\begin{aligned} &= \langle A^1 \partial_1 + A^2 \partial_2 + \dots + A^n \partial_n, B^1 \partial_1 + B^2 \partial_2 + \dots + B^n \partial_n \rangle \\ &\vdots \\ &= \underbrace{\langle A^1 \partial_1, B^1 \partial_1 \rangle + \langle A^1 \partial_1, B^2 \partial_2 \rangle + \dots}_{\rightarrow A^1 B^1 \langle \partial_1, \partial_1 \rangle} \quad \rightarrow A^1 B^2 \langle \partial_1, \partial_2 \rangle \end{aligned}$$

$$= \sum_i \sum_j A^i B_j g_{ij}, \quad \text{where } g_{ij} = \underbrace{\langle \partial_i, \partial_j \rangle}$$

Think dot product. Inner product generalizes the dot product.

For a statistical model $p(x, \theta)$, the manifold $S = \{p(x, \theta)\}$, the tangent vectors that constitute the basis are

$$\partial_i l(x, \theta), \text{ where } l(x, \theta) = \ln p(x, \theta).$$

$\xrightarrow{\text{red}} \frac{\partial}{\partial \theta^i} \ln p(x, \theta) = \frac{1}{p(x, \theta)} \frac{\partial p(x, \theta)}{\partial \theta^i}$

The vector space is

$$T_{\theta}^{(i)} = \{A(x) : A(x) = \sum A^i \partial_i l(x, \theta)\}$$

Here x is a R.V. Thus $T_{\theta}^{(i)}$ is the linear space spanned by r.v.s $\partial_i l(x, \theta)$

Any vector in the vector space (the tangent plane) can be expressed in terms of the vectors $\partial_i l(x, \theta)$. A^i are the components along $\partial_i l(x, \theta)$.

$$\partial_i \in T_\theta \iff \partial_i l(x, \theta) \in T_\theta^{(1)}$$

$T_\theta^{(1)}$ is called the 1-representation of the tangent space.

Consider

$$\ln p(x, \theta)$$

$$A(x) = \sum_{i=1}^n A^i \underbrace{\partial_i \ln p(x, \theta)}$$

$$E[A(x)] =$$

$$E \left[\sum_{i=1}^n A^i \frac{\partial}{\partial \theta^i} \ln p(x, \theta) \right]$$

$$= \sum_{i=1}^n A^i E \left[\frac{\partial}{\partial \theta^i} \ln p(x, \theta) \right]$$

$$= \sum_{i=1}^n A^i E \left[\frac{1}{p(x, \theta)} \frac{\partial p(x, \theta)}{\partial \theta^i} \right]$$

$$= \sum_{i=1}^n A^i \int_{\mathcal{X}} \left(\frac{1}{p(x, \theta)} \frac{\partial p(x, \theta)}{\partial \theta^i} \right) p(x, \theta) dx$$

$$= \sum_{i=1}^n A^i \int_{\mathcal{X}} \frac{\partial p(x, \theta)}{\partial \theta^i} dx$$

$$= \sum_{i=1}^n A^i \underbrace{\frac{\partial}{\partial \theta^i} \int_{\mathcal{X}} p(x, \theta) dx}_{=1}$$

$$= \sum_{i=1}^n A^i (0) = 0.$$

$$\boxed{E[g(x)] = \int_{\mathcal{X}} g(x) f_X(x) dx}$$

Riemannian Metric & Fisher Information

When the inner product $\langle A, B \rangle$ of two tangent vectors A and B is defined, the manifold is called a Riemannian space.

When we consider the 1-representations of A and B , respectively $A(x)$ and $B(x)$, the inner product is

$$\underbrace{\langle A(x), B(x) \rangle}_{\uparrow} = \underbrace{E[\underbrace{A(x)}_{\uparrow \text{ Zufall}} \underbrace{B(x)}_{\uparrow \text{ Zufall}}]}_{\text{Covarianz}} = \underbrace{\text{Cov}[A(x), B(x)]}_{\text{Covarianz}} \stackrel{\Delta}{=} E[A(x) B(x)] - E[A(x)] E[B(x)]$$

Note that $E[A(x)] = E[B(x)] = 0$ as we showed earlier.

Riemannian Metric & Fisher Information

$$A(x) = \sum_i A^i \partial_i \ln p(x, \theta) \quad \checkmark$$

$$B(x) = \sum_j B^j \partial_j \ln p(x, \theta) \quad \checkmark$$

Show that:

$$E[A(x) B(x)] = - \sum_i \sum_j A^i B^j \underbrace{E\left[\frac{\partial}{\partial \theta^i} \frac{\partial}{\partial \theta^j} \ln p(x, \theta)\right]}$$

(1) Expand $\underbrace{E\left[\frac{\partial}{\partial \theta^i} \frac{\partial}{\partial \theta^j} \ln p(x, \theta)\right]}$

(2) $\nearrow E\left[\left(\frac{\partial}{\partial \theta^i} \ln p(x, \theta)\right) \left(\frac{\partial}{\partial \theta^j} \ln p(x, \theta)\right)\right] \quad \checkmark$

$$E[A(x) B(x)] = E\left[\sum_i \sum_j \underbrace{A^i}_{\uparrow} \underbrace{B^j}_{\uparrow} \left(\frac{\partial}{\partial \theta^i} \ln p(x, \theta)\right) \left(\frac{\partial}{\partial \theta^j} \ln p(x, \theta)\right)\right]$$

Fisher Information Matrix

$$\langle \partial_i, \partial_j \rangle = g_{ij}(\theta) = \underline{E[\partial_i l(x, \theta) \partial_j l(x, \theta)]}$$

- Can also be written as

$$\begin{aligned} \langle \partial_i, \partial_j \rangle &= g_{ij}(\theta) = -E[\partial_i \partial_j l(x, \theta)] \\ &= -E \left[\frac{\partial}{\partial \theta_i} \frac{\partial}{\partial \theta_j} \ln p(x, \theta) \right] \end{aligned}$$

NATURAL GRADIENT

$$\phi(p) \rightarrow \theta \text{ in } \mathbb{R}^n$$

$$\text{Curve } c(t) \mapsto \theta(t)$$

$$f(\theta(t+\Delta t)) \approx f(\theta(t)) + \dot{c}(t) \Delta t$$

If f is a loss function, we would want to choose $\dot{c}(t)$ that minimizes $f(\theta(t+\Delta t))$

Any $\dot{c}(t)$ can be written in terms of the basis.

$$\text{Any } \dot{c}(t) = \nabla_{\theta} f(\theta(t))^T \begin{bmatrix} A_i \end{bmatrix}$$

$$\left. \begin{array}{l} \text{I want to minimize } \nabla_{\theta} f^T \vec{a} \\ \text{s.t. } |\vec{a}|^2 = 1. \end{array} \right\} \vec{a} \triangleq \begin{bmatrix} A_i \end{bmatrix}$$

That is, I want to minimize

$$\nabla_{\theta} f^T \vec{a}, \text{ such that}$$

$$\sum_i \sum_j A_i A_j g_{ij}(\theta) = 1$$

At the local minimum, we must satisfy:

$$\frac{\partial}{\partial A_i} \left[\nabla_{\theta} f^T \vec{a} - \lambda \left(\sum_i \sum_j A_i A_j g_{ij}(\theta) \right) \right] = 0$$

$$\frac{\partial}{\partial A_i} \left(\nabla_{\theta} f^T \vec{a} - \lambda \vec{a}^T G \vec{a} \right) = 0, i=1, \dots, n$$

Fisher Info Matrix.

$$\text{Solving gives } \vec{a} = \frac{1}{2\lambda} G^{-1} \nabla_{\theta} f(\theta)$$

Natural Gradient is $G^{-1}(\theta) \nabla_{\theta} f(\theta)$

A NATURAL POLICY GRADIENT

By Sham Kakade.

Key Takeaways:

→ Natural gradient based updates move toward choosing optimal greedy actions (as would result from the policy improvement step)

Outline:

(*) Define average cost. We will assume a continuing task and an undiscounted setting.

→ Average reward $\eta(\theta)$. θ is the parameter vector.

→ η is the function that we want to maximize over the manifold embedded by the PDF(s) parametrized by θ (and choice of approximation architecture).

(*) Define the natural gradient

$\nabla \eta(\theta)$ using the Fisher information matrix.

(*) [This is key] Show that the parameter update is locally moving in the direction of the policy that results from the policy improvement step.

→ Policy improvement (to guarantee improvement) assumes knowledge of the exact state/action-value function.

→ Exact value function isn't known.

→ Paper picks a "Compatible Function Approximation" (for a fair comparison)

↳ The approximation has the property that its natural gradient is equal to the natural gradient calculated assuming the exact value function.

} Follows from equations (3)-(5) of Thm 2 of the paper "Policy Gradient Method for Reinforcement Learning with Function Approximation"

By Sutton et al, in NIPS 2000.

Average Reward:

$$\eta(\pi) = \sum_{s,a} \underbrace{p^\pi(s)} \underbrace{\pi(a|s)} \underbrace{R(s,a)}$$

$$\nabla \eta(\theta) = \sum_{s,a} p^\pi(s) \nabla \pi(a|s, \theta) \theta^\pi(s,a)$$

↓
(policy
parameterization)

[Gradient is calculated
in the book by Sutton.
See policy gradient
theorem for continuing
task/average rewards]

[Can we calculate the above?]

Define the natural gradient:

For any state s , the set of $\pi(a|s, \theta)$ is
a manifold. $p(x, \theta) = \pi(x|s, \theta)$

For the manifold

$$g_{ij}(\theta) = \langle \partial_i, \partial_j \rangle = E \left[\frac{\partial}{\partial \theta_i} \ln \pi(a|s, \theta) \frac{\partial}{\partial \theta_j} \ln \pi(a|s, \theta) \right]$$

↑
[averaging over what]

Define $F_s(\theta) \triangleq \left[g_{ij}(\theta) \right]_{\substack{i=1, \dots, n \\ j=1, \dots, n}}$

Natural gradient is:

$$(F_s(\theta))^{-1} \nabla \eta(\theta)$$

↑

We will average over states s .

Steepest direction of descent is

$$\underline{F(\theta)^{-1} \nabla \eta(\theta)},$$

$$\underline{F(\theta)} = \underline{E_s[F_s(\theta)]}$$